

Digital Forensics Laboratory

EXPERIMENT NO: 02

DATE: August 9, 2026

TITLE: Recover deleted or damaged files from a storage device using TestDisk

1. AIM

To recover deleted files and repair a missing or corrupted partition from a storage device using the TestDisk data recovery command-line utility.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tool:** TestDisk (v7.3-WIP Data Recovery Utility)
- **Hardware:** Target Storage Device (e.g., hp v221w 16GB USB Device).

3. THEORY

TestDisk is a powerful, open-source data recovery software primarily designed to help recover lost partitions and make non-booting disks bootable again when these symptoms are caused by faulty software, certain types of viruses, or human error (such as accidentally deleting a Partition Table). It can also be utilized to undelete files from FAT, exFAT, NTFS, and ext2 file systems. TestDisk queries the BIOS or the OS in order to find the hard drives and their characteristics, performs a quick check of the disk structure, and looks for lost partitions or deleted files.

4. PROCEDURE

Step 1: Log Creation & Disk Selection

1. Open the testdisk_win.exe application as an Administrator.
2. **Log Creation:** Choose [Create] to instruct TestDisk to create a log file containing technical information and messages. Press Enter to proceed.



```
C:\Users\tamil\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
>[ Create ] Create a new log file
  [ Append ] Append information to log file
  [ No Log ] Don't record anything
```

Figure 1: Initial TestDisk dashboard prompting for log creation

3. **Disk Selection:** All connected hard drives will be detected and listed. Use the up/down arrow keys to select the target hard drive containing the lost partitions or deleted files (e.g., Disk \\.\PhysicalDrive1 - 16 GB / 15 GiB - hp v221w). Press Enter to Proceed.

```
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TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - EVM25/512GB
>Disk \\.\PhysicalDrive1 - 16 GB / 15 GiB - hp v221w

>[Proceed ] [ Quit ]

Note: Serial number 0404090000008818
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

Figure 2: Selecting the physical storage drive in TestDisk

Step 2: Partition Table Type Selection

1. TestDisk displays the available partition table types.
2. Select the partition table type. Usually, the default value is the correct one as TestDisk auto-detects the partition table type (e.g., [Intel] Intel/PC partition). Press Enter to Proceed.

```
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Disk \\.\PhysicalDrive1 - 16 GB / 15 GiB - hp v221w

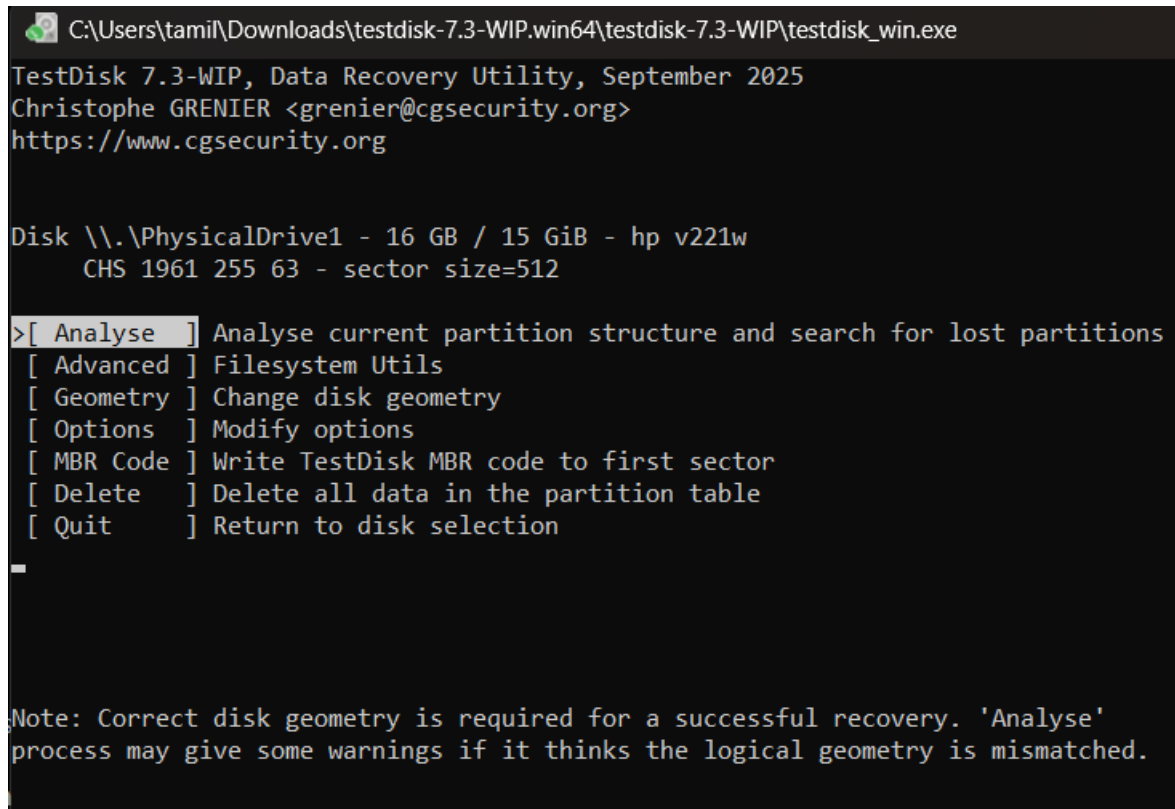
Please select the partition table type, press Enter when done.
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

Figure 3: Selecting the Intel/PC partition table type

Step 3: Current Partition Table Status

1. TestDisk displays the main menu options. Use the default menu [Analyse] to check your current partition structure and search for lost partitions. Confirm at Analyse with Enter.



```
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Disk \\.\PhysicalDrive1 - 16 GB / 15 GiB - hp v221w
  CHS 1961 255 63 - sector size=512

>[ Analyse ] Analyse current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options  ] Modify options
[ MBR Code ] Write TestDisk MBR code to first sector
[ Delete   ] Delete all data in the partition table
[ Quit     ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

Figure 4: Choosing the Analyse option to examine the partition structure

2. The current partition structure is listed. Examine the structure for missing partitions and errors (e.g., Invalid NTFS boot, missing logical partitions). Confirm at [Quick Search] to proceed.

```

C:\Users\tamil\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
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https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 16 GB / 15 GiB - CHS 1961 255 63
Current partition structure:
  Partition          Start          End      Size in sectors
  1 P HPFS - NTFS      3 48 30   1961 47 5    31455183 [INBA]
No partition is bootable

*=Primary bootable  P=Primary  L=Logical  E=Extended  D=Deleted
>[Quick Search]  [ Backup ]
Try to locate partition. Press Enter

```

Figure 5: Initiating a Quick Search for lost partitions

Step 4: Deeper Search (If Required)

1. If the quick search does not find the required partition or if partitions overlap (status 'D' for deleted), highlight the [Deeper Search] menu and press Enter.
2. Deeper Search will scan each cylinder for FAT32 backup boot sectors, NTFS backup boot superblocks, etc., to detect more partitions.

```

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TestDisk 7.3-WIP, Data Recovery Utility, September 2025
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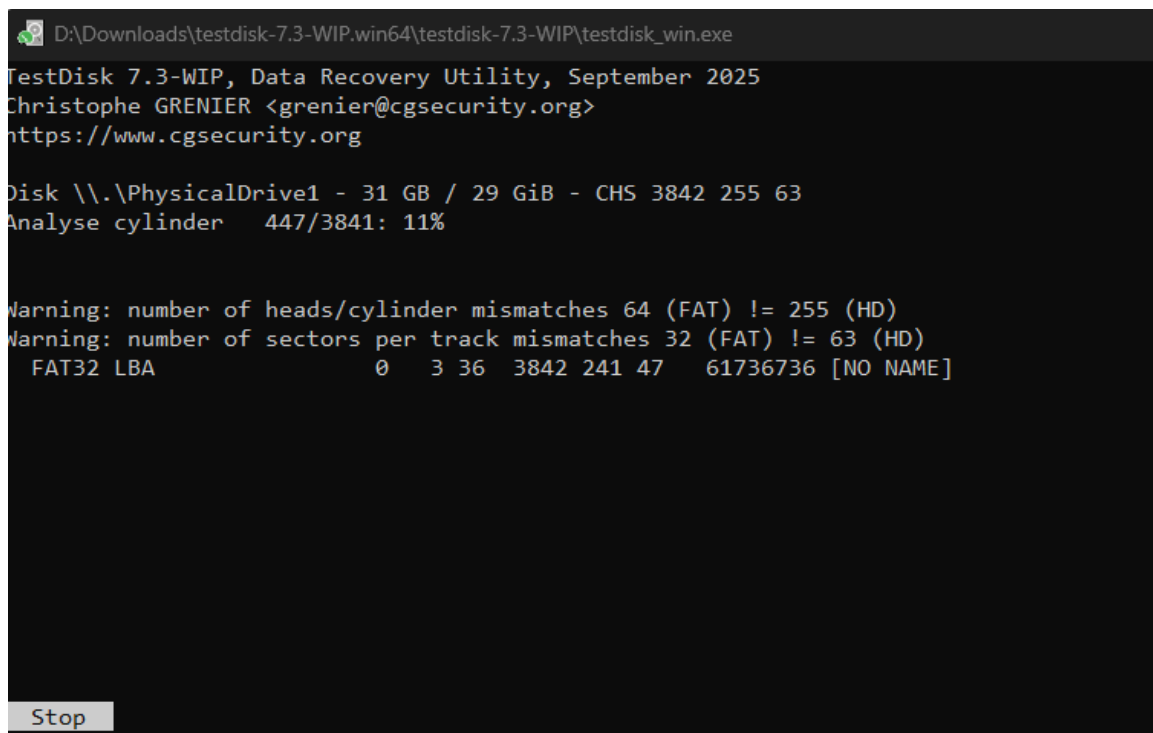
Disk \\.\PhysicalDrive1 - 16 GB / 15 GiB - CHS 1961 255 63

  Partition          Start          End      Size in sectors
  1 E extended LBA      3  0  1   1961 47 37    31458268
  5 L FAT32 LBA         3 48 62   1961 47 37    31455183

[ Quit ] [ Return ] >[Deeper Search] [ Write ] [Extd Part]
Try to find more partitions.

```

Figure 6: Selecting the Deeper Search option

A screenshot of the TestDisk 7.3-WIP application window. The title bar shows the file path 'D:\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe'. The main text area displays the following information: 'TestDisk 7.3-WIP, Data Recovery Utility, September 2025', 'Christophe GRENIER <grenier@cgsecurity.org>', 'https://www.cgsecurity.org', 'Disk \\.\PhysicalDrive1 - 31 GB / 29 GiB - CHS 3842 255 63', 'Analyse cylinder 447/3841: 11%', 'Warning: number of heads/cylinder mismatches 64 (FAT) != 255 (HD)', 'Warning: number of sectors per track mismatches 32 (FAT) != 63 (HD)', and a table with the header 'FAT32 LBA' and one row of values: '0 3 36 3842 241 47 61736736 [NO NAME]'. At the bottom left, there is a 'Stop' button.

```
D:\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 31 GB / 29 GiB - CHS 3842 255 63
Analyse cylinder 447/3841: 11%

Warning: number of heads/cylinder mismatches 64 (FAT) != 255 (HD)
Warning: number of sectors per track mismatches 32 (FAT) != 63 (HD)
FAT32 LBA          0  3 36 3842 241 47 61736736 [NO NAME]
```

Figure 7: Deeper Search actively scanning disk cylinders

Step 5: Identifying and Accessing the Correct Partition

1. Once the search is complete, highlight the correct partition you wish to recover or extract files from (e.g., >L FAT32 LBA).
2. Press the p key to list the files inside that partition.

```
D:\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
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https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 31 GB / 29 GiB - CHS 3842 255 63
Partition      Start      End      Size in sectors
>L FAT32 LBA    0  3 36  3842 241 47  61736736 [NO NAME]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
FAT32, blocksize=16384, 31 GB / 29 GiB
```

Figure 8: Highlighting the located FAT32 partition

Step 6: Recovering / Copying Deleted Files

1. In the file listing, navigate through the folders using arrow keys. Files listed in red indicate deleted entries.
2. Highlight the specific deleted file you want to recover (e.g., images (1).jpg).

```
TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
  HPFS - NTFS          3  48 30  1961  47  5   31455183 [INBA]
Directory /

dr-xr-xr-x   0   0   0  9-Aug-2026 17:34 .
dr-xr-xr-x   0   0   0  9-Aug-2026 17:34 ..
dr-xr-xr-x   0   0   0 12-Dec-2025 22:35 System Volume Information
-r--r--r--   0   0 564127 2-Aug-2026 19:33 ITLAB7 (1).jpeg
-r--r--r--   0   0   133 2-Aug-2026 19:33 ITLAB7 (1).jpeg:Zone.Identifier
-r--r--r--   0   0 564127 2-Aug-2026 19:32 ITLAB7.jpeg
-r--r--r--   0   0   133 2-Aug-2026 19:32 ITLAB7.jpeg:Zone.Identifier
-r--r--r--   0   0 23509 2-Aug-2026 19:34 images (1).jpg
>-r--r--r--   0   0   189 2-Aug-2026 19:34 images (1).jpg:Zone.Identifier
-r--r--r--   0   0 24143 2-Aug-2026 19:25 images.jpg
-r--r--r--   0   0   188 2-Aug-2026 19:25 images.jpg:Zone.Identifier
-r--r--r--   0   0 73297 2-Aug-2026 19:24 pic-blog-1.jpg
-r--r--r--   0   0   125 2-Aug-2026 19:24 pic-blog-1.jpg:Zone.Identifier

                                Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file
```

Figure 9: Listing the directory contents and selecting a file for recovery

3. Press c to copy the current selected file (or C to copy all selected files).
4. You will be prompted to select a destination directory to save the recovered file.
Navigate to the desired local folder and press C when the destination is correct.

```

TestDisk 7.3-WIP, Data Recovery Utility, September 2025

Please select a destination where the marked files will be copied.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory C:\Users\tamil\Downloads\testdisk-7.3-WIP.win64\testdisk-7.3-WIP
>drwx----- 197609 197609      0 9-Aug-2026 21:28 .
drwx----- 197609 197609      0 9-Aug-2026 19:48 ..
drwx----- 197609 197609      0 9-Aug-2026 19:48 63
drwx----- 197609 197609      0 9-Aug-2026 21:30 Rec
drwx----- 197609 197609      0 9-Aug-2026 19:48 platforms
-rwx----- 197609 197609    220 9-Aug-2026 19:48 AUTHORS.txt
-rwx----- 197609 197609   18326 9-Aug-2026 19:48 COPYING.txt
-rwx----- 197609 197609    120 9-Aug-2026 19:48 INFO
-rwx----- 197609 197609   20989 9-Aug-2026 19:48 NEWS.txt
-rwx----- 197609 197609  6272168 9-Aug-2026 19:48 Qt5Core.dll
-rwx----- 197609 197609  6897224 9-Aug-2026 19:48 Qt5Gui.dll
-rwx----- 197609 197609  7271983 9-Aug-2026 19:48 Qt5Widgets.dll
-rwx----- 197609 197609    350 9-Aug-2026 19:48 THANKS.txt
-rwx----- 197609 197609    40 9-Aug-2026 19:48 VERSION.txt
-rwx----- 197609 197609  5457751 9-Aug-2026 19:48 cygwf-2.dll
-rwx----- 197609 197609   708066 9-Aug-2026 19:48 cyggcc_s-seh-1.dll
-rwx----- 197609 197609  1380305 9-Aug-2026 19:48 cygiconv-2.dll
-rwx----- 197609 197609  2238354 9-Aug-2026 19:48 cygjpeg-8.dll
-rwx----- 197609 197609  3785114 9-Aug-2026 19:48 cygncursesw-10.dll
-rwx----- 197609 197609  3549700 9-Aug-2026 19:48 cygwin1.dll
-rwx----- 197609 197609   306623 9-Aug-2026 19:48 cygz.dll
-rwx----- 197609 197609    504 9-Aug-2026 19:48 documentation.html
Next

```

Figure 10: Selecting the destination folder for the recovered file

5. TestDisk will display a confirmation message indicating a successful recovery (e.g., "Copy done! 1 ok, 0 failed").

```

TestDisk 7.3-WIP, Data Recovery Utility, September 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
HPFS - NTFS      3  48 30  1961  47  5  31455183 [INBA]
Directory /images.jpg
Copy done! 1 ok, 0 failed
dr-xr-xr-x  0  0      0 9-Aug-2026 17:34 .
dr-xr-xr-x  0  0      0 9-Aug-2026 17:34 ..
dr-xr-xr-x  0  0      0 12-Dec-2025 22:35 System Volume Information
-r--r--r--  0  0   564127 2-Aug-2026 19:33 ITLAB7 (1).jpeg
-r--r--r--  0  0    133 2-Aug-2026 19:33 ITLAB7 (1).jpeg:Zone.Identifier
-r--r--r--  0  0   564127 2-Aug-2026 19:32 ITLAB7.jpeg
-r--r--r--  0  0    133 2-Aug-2026 19:32 ITLAB7.jpeg:Zone.Identifier
-r--r--r--  0  0   23509 2-Aug-2026 19:34 images (1).jpg
-r--r--r--  0  0    189 2-Aug-2026 19:34 images (1).jpg:Zone.Identifier
-r--r--r--  0  0   24143 2-Aug-2026 19:25 images.jpg
-r--r--r--  0  0    188 2-Aug-2026 19:25 images.jpg:Zone.Identifier
-r--r--r--  0  0   73297 2-Aug-2026 19:24 pic-blog-1.jpg
-r--r--r--  0  0    125 2-Aug-2026 19:24 pic-blog-1.jpg:Zone.Identifier

                                Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file.

```

Figure 11: Successful copy/recovery of the deleted file

Step 7: Partition Table and Boot Sector Recovery (Optional)

1. Press q to quit the file listing and return to the partition selection.
2. Use the left/right arrow keys to change the status of a partition from D (Deleted) to L (Logical) or P (Primary) to recover the partition itself. Press Enter.

3. If all partitions are correctly listed, confirm at [Write] with Enter, press y to confirm, and then OK.
4. If a boot sector is damaged, use the [Backup BS] option to copy the backup boot sector over the damaged one.
5. Exit TestDisk and reboot the computer to access the restored partition data.

5. RESULT

The TestDisk data recovery utility was successfully executed to analyze the physical drive's geometry and partition tables. Lost partitions were successfully identified through deeper searching, and deleted files were effectively navigated, selected, and copied (recovered) to a secure local destination.